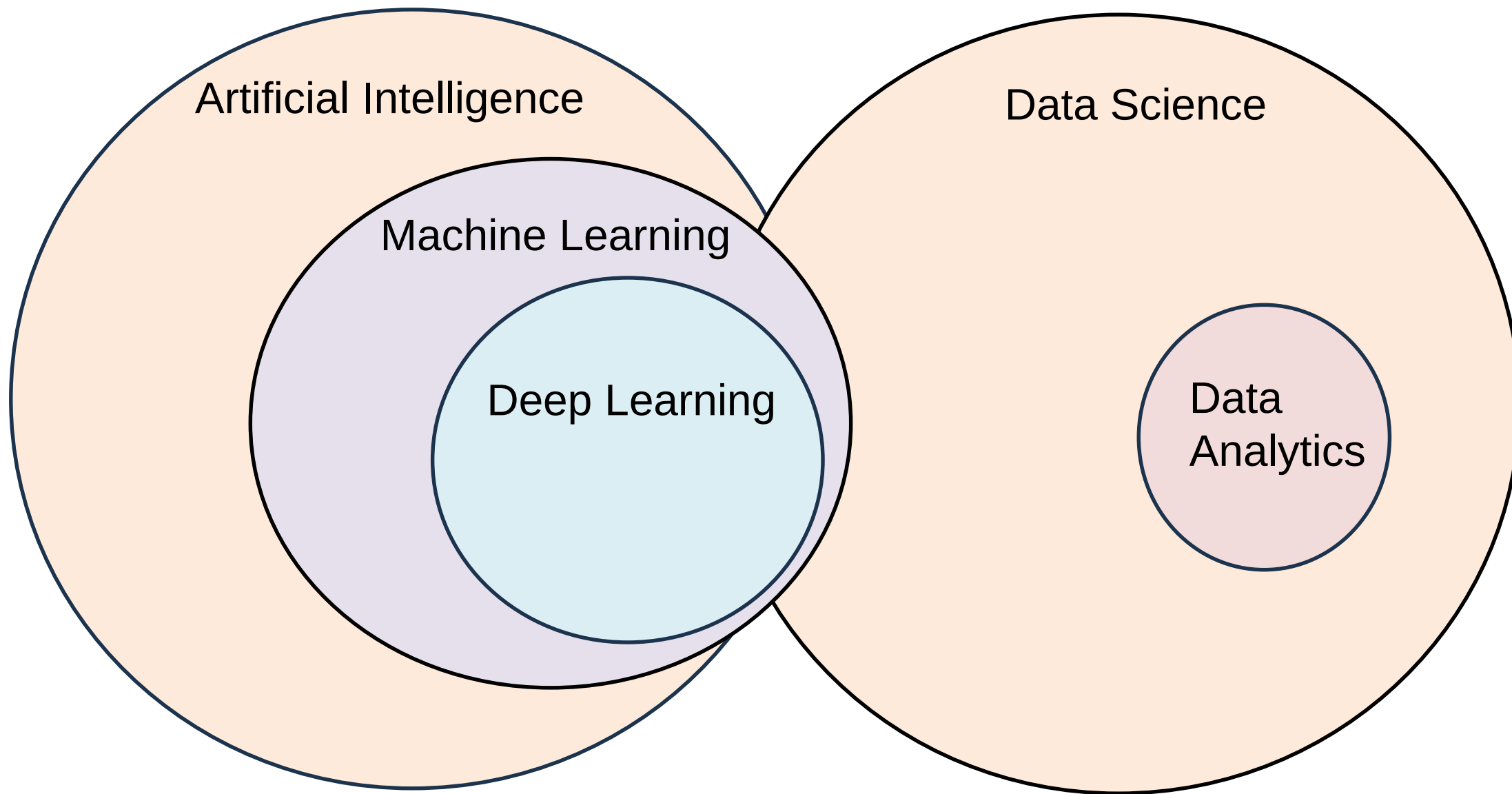


Buzz Words



Definitions

Artificial Intelligence

“The study of [intelligent] agents that receive precepts from the environment and take action” ~ Russell and Norvig

Data Science

“Data science is the study of data to extract meaningful insights for business” ~ AWS

Machine learning

“Machine learning is the ability to solve a problem without being explicitly programmed” ~Arthur Samuel

Deep Learning

“Deep learning is a form of machine learning that enables computers to learn from experience and understand the world in terms of a hierarchy of concepts” ~ Ian Goodfellow

Data Analytics

“Data analysis is the process of inspecting, [cleansing](#), [transforming](#), and [modeling data](#) with the goal of discovering useful information, informing conclusions, and supporting decision-making.” ~ [Meta S. Brown](#)

Data Science

- Statistical tools, methods and technology to generate meaning from data
- Use models and methods used in data analysis and apply this to new data
- Scope of DS
 - Data collection and preprocessing.
 - Applying appropriate models and algorithms to the data to answer these questions.
 - Interpreting the results.

Artificial Intelligence

- Solve cognitive problems with human intelligence
- Use applied data science technique and algorithms to compose and run complex machine based systems that approximate human intelligence
- Scope
 - Exploratory data analysis.
 - Dividing the task into algorithmic components to form a system.
 - Gathering test data to review and refine the suitability of the logical flow and complexity of the system.
 - Testing the system.
- Ex:
 - Learning
 - Human like expressions
 - Pattern recognition

DS

K Mean Clustering

Principle Component Analysis

Singular Value Decomposition

Apriori Algorithm

Independent Component Analysis

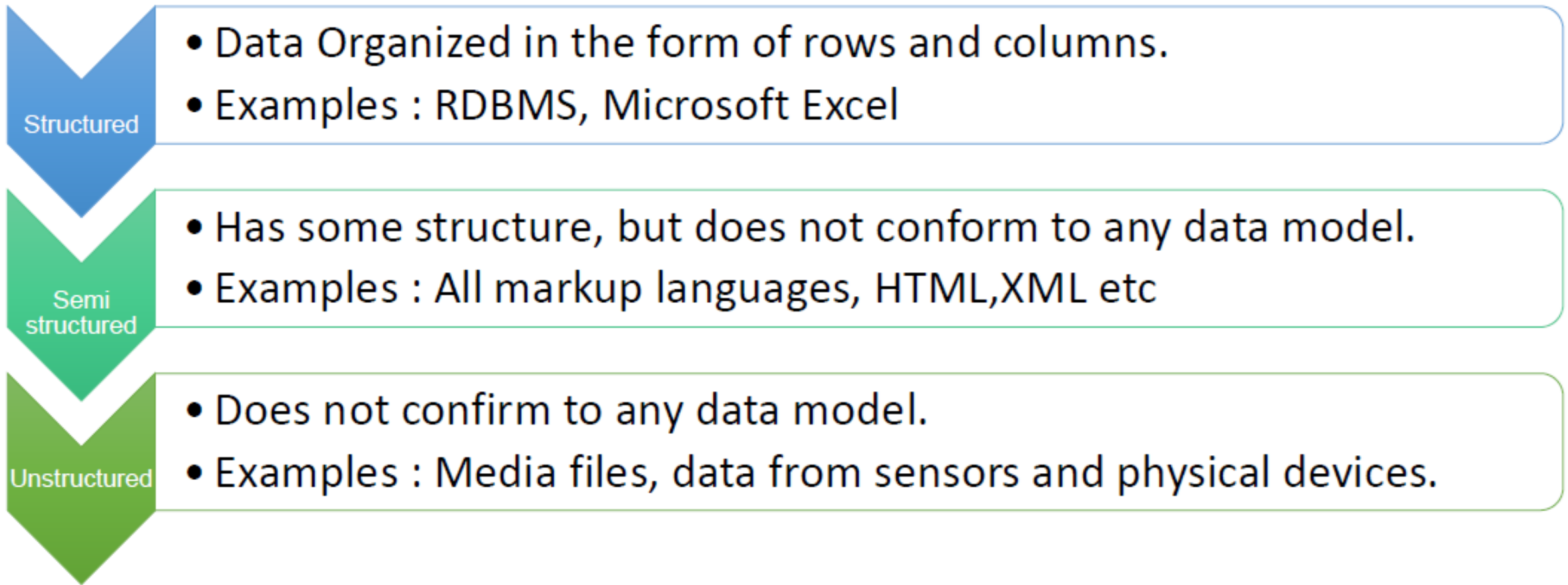
AI

Neural Network

Search and Optimization algorithm

Digital Data

Types of Digital Data



Structured

- Data Organized in the form of rows and columns.
- Examples : RDBMS, Microsoft Excel

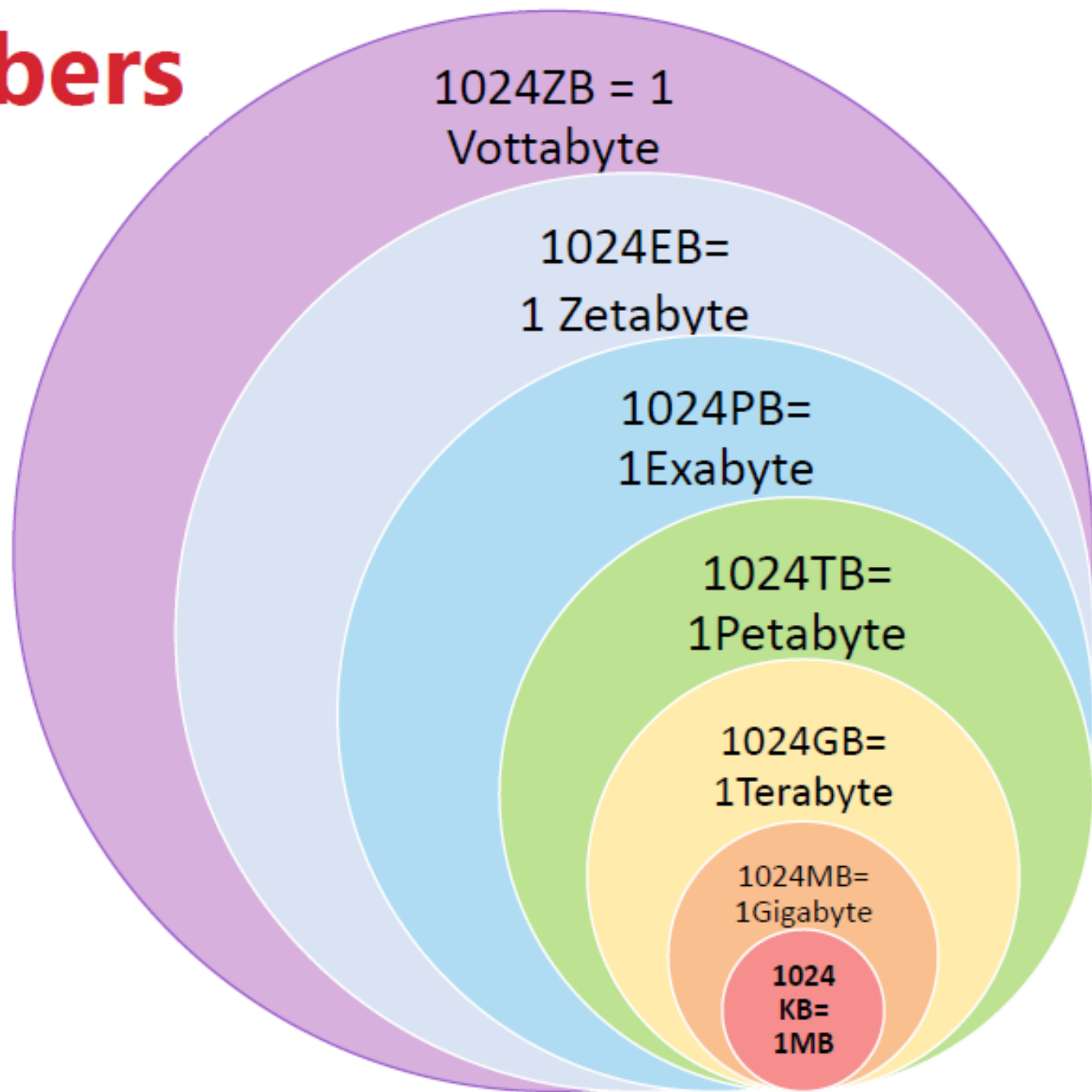
Semi structured

- Has some structure, but does not conform to any data model.
- Examples : All markup languages, HTML,XML etc

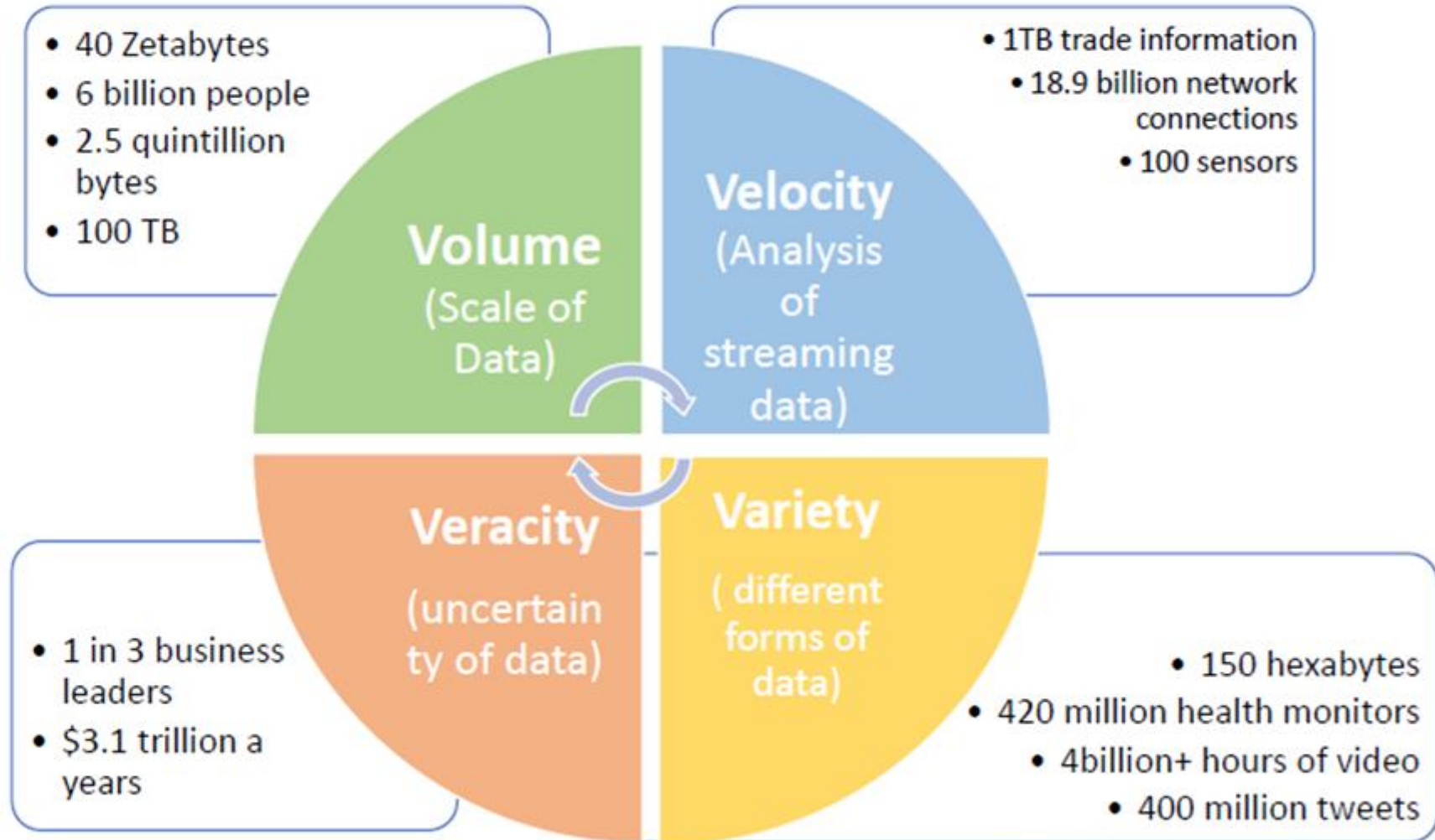
Unstructured

- Does not confirm to any data model.
- Examples : Media files, data from sensors and physical devices.

Big Data Numbers



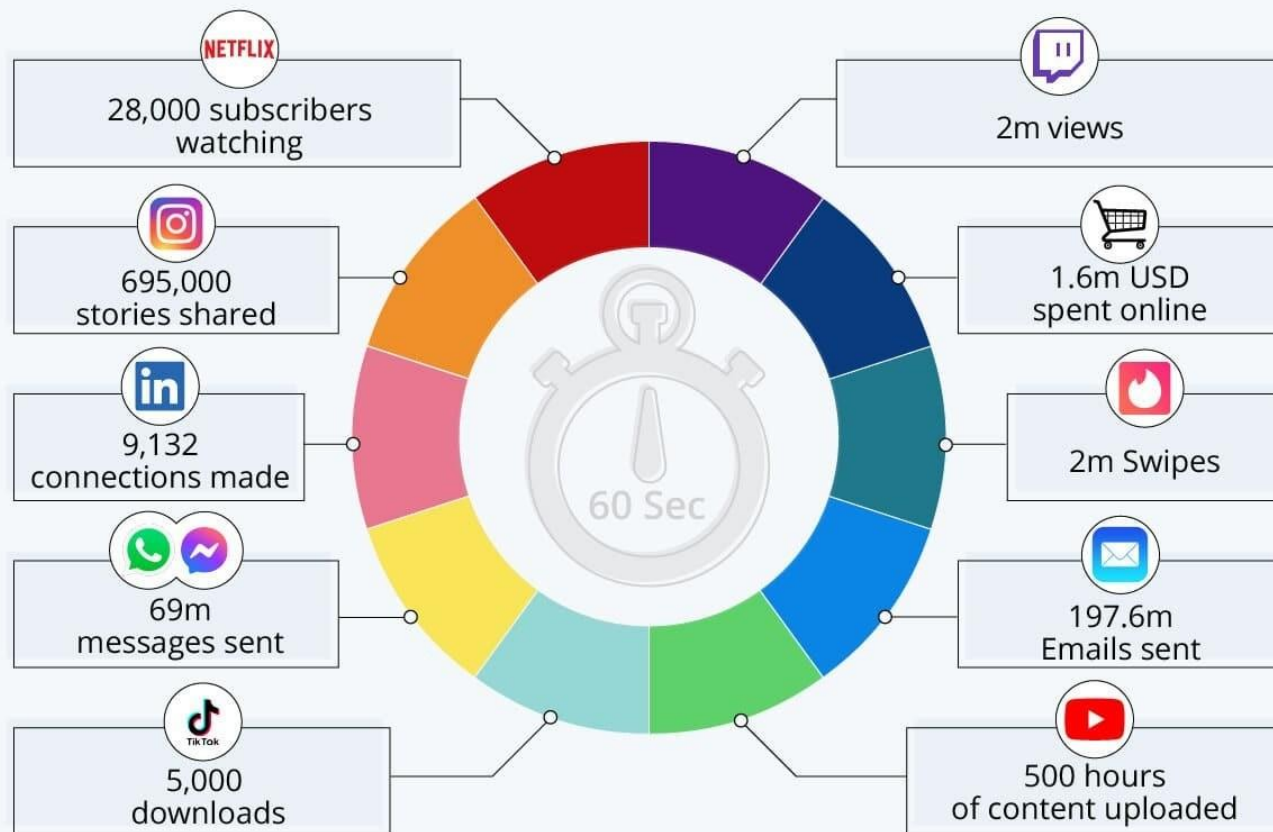
What is Big Data?



What Happens in 60 seconds over internet in 2023?

A Minute on the Internet in 2021

Estimated amount of data created
on the internet in one minute



Source: Lori Lewis via AllAccess



statista

Machine Learning

SL No	No. of Action Scene	No. of Comedy Scene	Class/ Label/ Category
1	100	15	Action
2	20	95	Comedy
3	90	5	Action
4	10	85	Comedy
5	50	50	

What is the category of the movie?



How many groups are there?

k nearest neighbour

Train Set

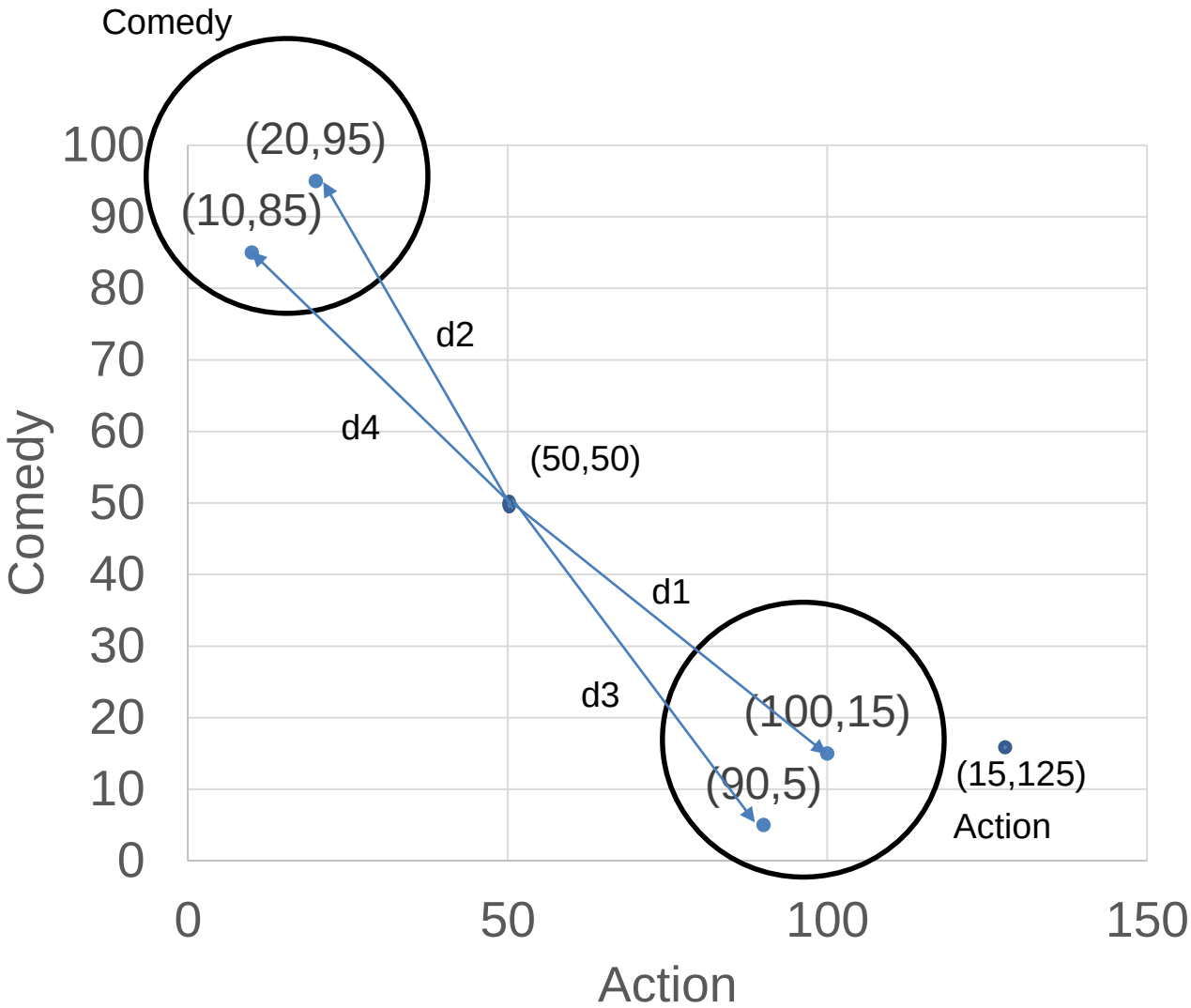
SL No	No. of Action Scene	No. of Comedy Scene	Class/ Labels/ Category
T1	100	15	Action
T2	20	95	Comedy
T3	90	5	Action
T4	10	85	Comedy

Test Set

T5	50	50	?
----	----	----	---

Steps to be followed

1. Compute Euclidian Distance from val/test to all the training data points
 2. Sort the data in the ascending order
 3. Compute the value of k
 - a. Use Validation data to determine the value of k. Voting scheme may be used for taking the majority voting.
 - b. Compute accuracy on different values of k and choose the one with the highest accuracy
 - c. Value of k should be odd in case of binary classification
- Use the value of k to compute the classification on test data and compute the accuracy

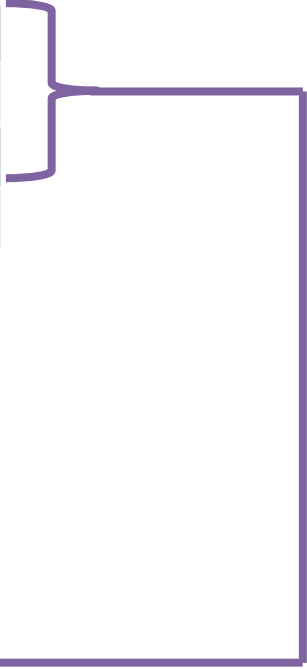


Step 1: Compute Euclidean Distance

Sl. No.	Distance	Train	Test	Class of Train	Distance
1	D1	T1	T5	Action	61.03277808
2	D2	T2	T5	Comedy	54.08326913
3	D3	T3	T5	Action	60.20797289
4	D4	T4	T5	Comedy	53.15072906

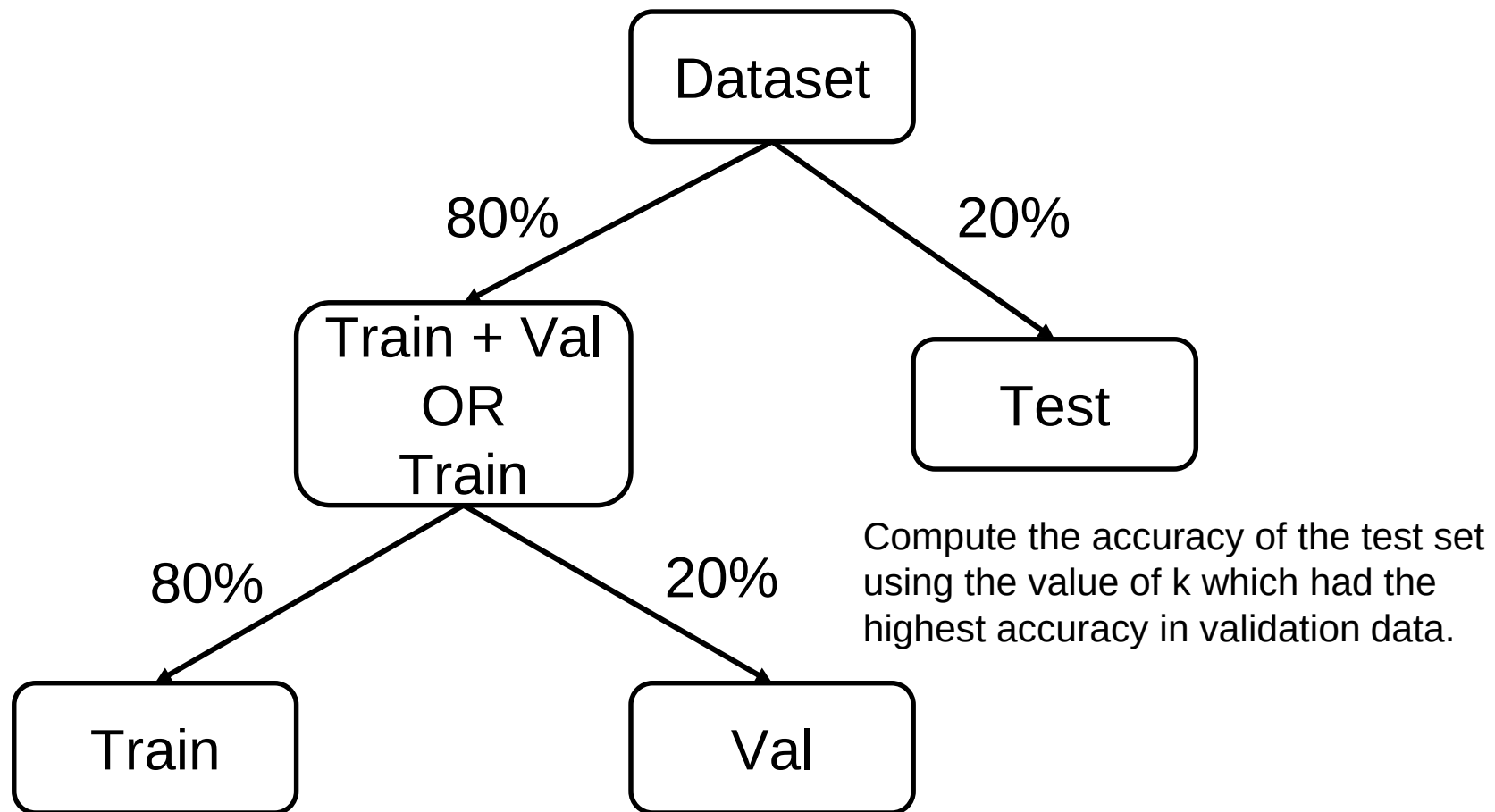
Step 2: Sort the data on the basis of Distance

Sl. No.	Distance	Train	Test	Class of Train	Distance
4	D4	T4	T5	Comedy	53.1507
2	D2	T2	T5	Comedy	54.0833
3	D3	T3	T5	Action	60.208
1	D1	T1	T5	Action	61.0328



Step 2: Determine the value of k

Different Values of k	Classes	Majority Vote
1	{Comedy}	Comedy
2	{Comedy, Comedy}	Comedy
3	{Comedy, Comedy, Action}	Comedy



Compute the accuracy of the test set using the value of k which had the highest accuracy in validation data.

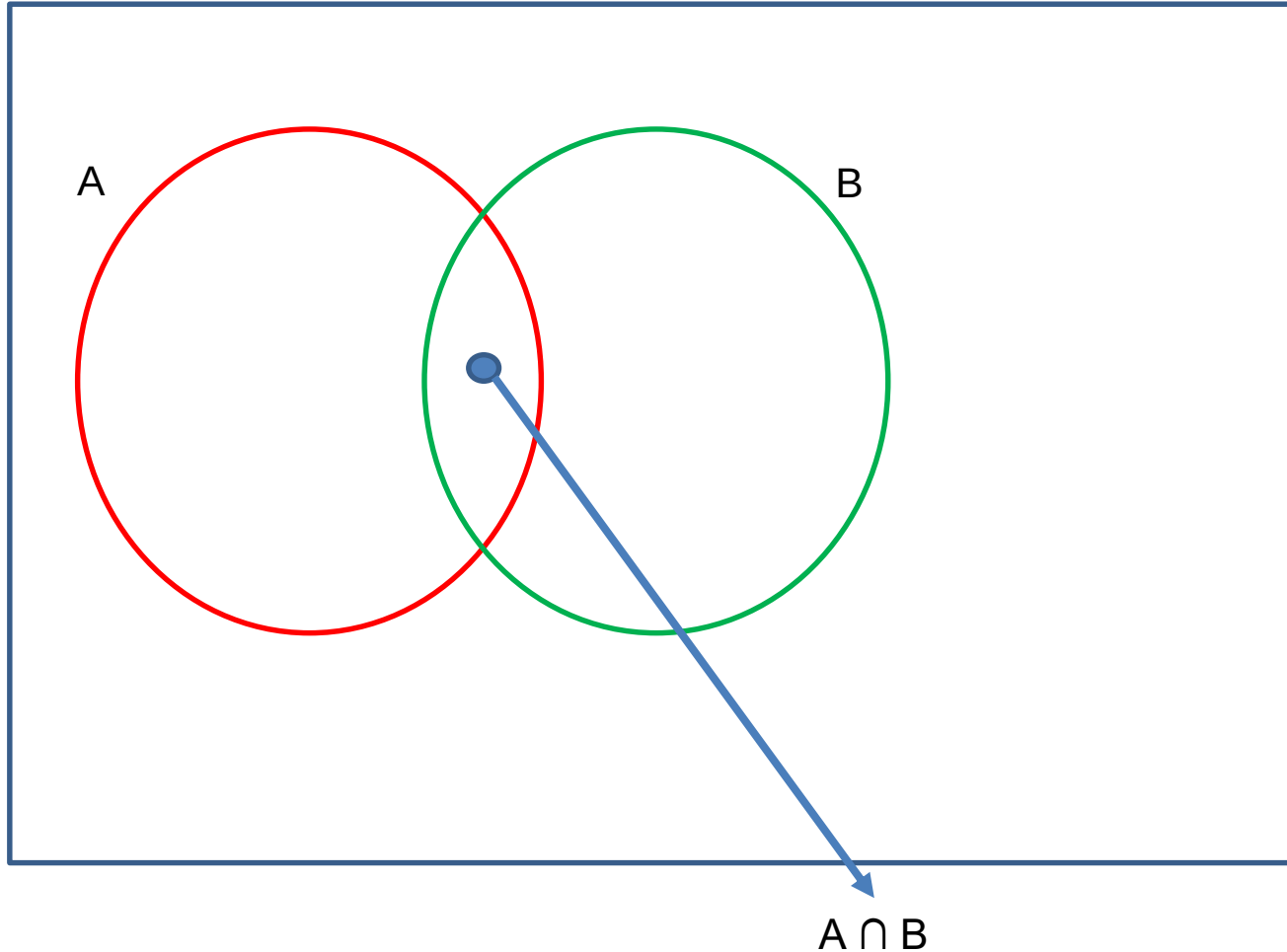
Compute Accuracy for Different Values of k using Validation Dataset, select the **k with the highest accuracy**

	Train (50)	Test(50)	Train (60)	Test(40)	Train (70)	Test(30)	Train (80)	Test(20)	Train (99.5)	Test(0.5)
100	50	50	60	40	70	30	80	20	99.5	0.5
1000	500	500	600	400	700	300	800	200	995	5
10,000	5000	5000	6000	4000	7000	3000	8000	2000	9950	50
1,00,000	50000	50000	60000	40000	70000	30000	80000	20000	99500	500
10,00,000	500000	500000	600000	400000	700000	300000	800000	200000	995000	5000
1,00,00,000	5000000	5000000	6000000	4000000	7000000	3000000	8000000	2000000	9950000	50000

Naive Bayes

Bayesian Framework

Universal Set S



Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \longrightarrow P(A \cap B) = P(A|B)P(B)$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} \longrightarrow P(B \cap A) = P(B|A)P(A)$$

$$A \cap B = B \cap A$$

$$P(B|A)P(A) = P(A|B)P(B)$$

Naïve Bayes Equation

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

$A = \{\theta_1, \theta_2, \theta_3 \dots \theta_n\}$; Set of Independent Events/Features
 $c_i = \text{Labels/Targets}$

$$P(c_i|\theta_1, \theta_2, \theta_3 \dots \theta_n) = \frac{P(\theta_1, \theta_2, \theta_3 \dots \theta_n|c_i)P(c_i)}{P(\theta_1, \theta_2, \theta_3 \dots \theta_n)}$$

$$P(c_i|\theta_1, \theta_2, \theta_3 \dots \theta_n) = \frac{P(\theta_1|c_i)P(\theta_2|c_i)P(\theta_3|c_i)\dots P(\theta_n|c_i) P(c_i)}{P(\theta_1)P(\theta_2)P(\theta_3) \dots P(\theta_n)}$$

$$P(c_i|\theta_1, \theta_2, \theta_3 \dots \theta_n) = P(\theta_1|c_i)P(\theta_2|c_i)P(\theta_3|c_i)\dots P(\theta_n|c_i) P(c_i)$$

$$\text{Predicted Class} = \underset{c_i}{\operatorname{argmax}} P(c_i) \prod_{i=1}^n P(\theta_i|c_i)$$

Naïve Bayes Classifier

Conditional Probability of Independent Events

$$P(\theta_1, \theta_2, \theta_3 \dots \theta_n|c_i) = P(\theta_1|c_i)P(\theta_2|c_i)P(\theta_3|c_i)\dots P(\theta_n|c_i)$$

$$P(\theta_1, \theta_2, \theta_3 \dots \theta_n) = P(\theta_1)P(\theta_2)P(\theta_3) \dots P(\theta_n)$$

We can Ignore the Denominator

- $P(c_i|\theta)$ is the posterior probability of class (c_i , target/labels) given predictor (θ , attributes)
- $P(c_i)$ is the prior probability of class
- $P(\theta|c_i)$ is the likelihood which is the probability of predictor given class
- $P(\theta)$ is the prior probability of predictor

Decision Tree: Information Gain

Sl. No.	Outlook	Temp	Humidity	Windy	Play Golf
1	Rainy	Hot	High	Weak	No
2	Rainy	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Sunny	Mild	High	Weak	Yes
5	Sunny	Cool	Normal	Weak	Yes
6	Sunny	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Rainy	Mild	High	Weak	No
9	Rainy	Cool	Normal	Weak	Yes
10	Sunny	Mild	Normal	Weak	Yes
11	Rainy	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Sunny	Mild	High	Strong	No

Outlook				
	Yes	No	P(Yes)	P(No)
Sunny	3	2	3/9	2/5
Overcast	4	0	4/9	0/5
Rainy	2	3	2/9	3/5
Total	9	5	100%	100%

Temperature				
	Yes	No	P(Yes)	P(No)
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
Cool	3	1	3/9	1/5
Total	9	5	100%	100%

Humidity				
	Yes	No	P(Yes)	P(No)
High	3	4	3/9	4/5
Normal	6	1	6/9	1/5
Total	9	5	100%	100%

Wind				
	Yes	No	P(Yes)	P(No)
Weak	6	2	6/9	2/5
Strong	3	3	3/9	3/5
Total	9	5	100%	100%

What is the probability that the match will be played “today” if it is Sunny, Cool, Normal, and Weak.

$$P(\text{Yes}) = \frac{\text{Number of Yes}}{\text{Total Number of Matches}} = \frac{9}{14} \quad P(\text{Sunny}|\text{Yes}) = \frac{3}{9} \quad P(\text{Cool}|\text{Yes}) = \frac{3}{9} \quad P(\text{Normal}|\text{Yes}) = \frac{6}{9} \quad P(\text{Weak}|\text{Yes}) = \frac{6}{9}$$

$$P(\text{No}) = \frac{\text{Number of No}}{\text{Total Number of Matches}} = \frac{5}{14} \quad P(\text{Sunny}|\text{No}) = \frac{2}{5} \quad P(\text{Cool}|\text{No}) = \frac{1}{5} \quad P(\text{Normal}|\text{No}) = \frac{1}{5} \quad P(\text{Weak}|\text{No}) = \frac{2}{5}$$

What is the probability that the match will be played if it is Sunny, Cool, Normal, and Weak.

$$P(\text{Yes}) = \frac{\text{Number of Yes}}{\text{Total Number of Matches}} = \frac{9}{14} \quad P(\text{Sunny}|\text{Yes}) = \frac{2}{9} \quad P(\text{Cool}|\text{Yes}) = \frac{3}{9} \quad P(\text{Normal}|\text{Yes}) = \frac{6}{9} \quad P(\text{Weak}|\text{Yes}) = \frac{6}{9}$$

$$P(\text{No}) = \frac{\text{Number of No}}{\text{Total Number of Matches}} = \frac{5}{14} \quad P(\text{Sunny}|\text{No}) = \frac{3}{5} \quad P(\text{Cool}|\text{No}) = \frac{1}{5} \quad P(\text{Normal}|\text{No}) = \frac{1}{5} \quad P(\text{Weak}|\text{No}) = \frac{2}{5}$$

$$P(\text{Yes}|\text{today}) = P(\text{Sunny}|\text{Yes}) P(\text{Cool}|\text{Yes}) P(\text{Normal}|\text{Yes}) P(\text{Weak}|\text{Yes}) P(\text{Yes}) = \frac{3}{9} \times \frac{3}{9} \times \frac{6}{9} \times \frac{6}{9} \times \frac{9}{14} = 0.0317$$

The Match will be played since the $P(\text{Yes}|\text{today}) > P(\text{No}|\text{today})$

$$P(\text{No}|\text{today}) = P(\text{Sunny}|\text{No}) P(\text{Cool}|\text{No}) P(\text{Normal}|\text{No}) P(\text{Weak}|\text{No}) P(\text{No}) = \frac{2}{5} \times \frac{1}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14} = 0.00228$$

$$P(\text{Yes}|\text{today}) + P(\text{No}|\text{today}) = 1$$

$$P(\text{Yes}|\text{today}) = \frac{0.0317}{0.0317 + 0.00228} = 0.86$$

$$P(\text{No}|\text{today}) = \frac{0.00228}{0.0317 + 0.00228} = 0.139$$

What is the probability that the match will be played if it is Sunny, Cool, Normal, and Weak.

$$P(\text{Yes}) = \frac{\text{Number of Yes}}{\text{Total Number of Matches}} = \frac{9}{14} \quad P(\text{Sunny}|\text{Yes}) = \frac{2}{9} \quad P(\text{Cool}|\text{Yes}) = \frac{3}{9} \quad P(\text{Normal}|\text{Yes}) = \frac{6}{9} \quad P(\text{Weak}|\text{Yes}) = \frac{6}{9}$$

$$P(\text{No}) = \frac{\text{Number of No}}{\text{Total Number of Matches}} = \frac{5}{14} \quad P(\text{Sunny}|\text{No}) = \frac{3}{5} \quad P(\text{Cool}|\text{No}) = \frac{1}{5} \quad P(\text{Normal}|\text{No}) = \frac{1}{5} \quad P(\text{Weak}|\text{No}) = \frac{2}{5}$$

$$P(\text{Yes}|\text{today}) = \frac{P(\text{Sunny}|\text{Yes}) P(\text{Cool}|\text{Yes}) P(\text{Normal}|\text{Yes}) P(\text{Weak}|\text{Yes}) P(\text{Yes})}{P(\text{today})} = \frac{\frac{2}{9} \times \frac{3}{9} \times \frac{6}{9} \times \frac{6}{9} \times \frac{9}{14}}{P(\text{today})} = \frac{0.021164}{P(\text{today})}$$

$$P(\text{No}|\text{today}) = \frac{P(\text{Sunny}|\text{No}) P(\text{Cool}|\text{No}) P(\text{Normal}|\text{No}) P(\text{Weak}|\text{No}) P(\text{No})}{P(\text{today})} = \frac{\frac{3}{5} \times \frac{1}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14}}{P(\text{today})} = \frac{0.003429}{P(\text{today})}$$

$$P(\text{Yes}|\text{today}) + P(\text{No}|\text{today}) = 1$$

$$P(\text{Yes}|\text{today}) = \frac{\frac{0.021164}{P(\text{today})}}{\frac{0.021164}{P(\text{today})} + \frac{0.003429}{P(\text{today})}} = 0.86$$

$$P(\text{No}|\text{today}) = \frac{\frac{0.003429}{P(\text{today})}}{\frac{0.021164}{P(\text{today})} + \frac{0.003429}{P(\text{today})}} = 0.139$$

The term $P(\text{today})$ appears in both the numerator and the denominator, so it will cancel out. There wont be an y change in the final results.

$$\text{Predicted Class} = \operatorname{argmax}_{c_i} P(c_i) \prod_{i=1}^n P(\theta_i|c_i)$$